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A mathematical and experimental study of ant foraging trail dynamics

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Abstract

In this article, we present a mathematical model coupled to an experimental study of ant foraging trails. Our laboratory experiments on *Tetramorium caespitum* do not find a strong relationship between ant densities and velocities, a common assumption in traffic modeling. Rather, we find that higher order effects play a major role in observed behavior, and our model reflects this by including inertial terms in the evolution equation. A linearization of the resulting system yields left- and right-moving waves, in agreement with laboratory measurements. The linearized system depends upon Froude numbers reflecting a ratio of the energy stored in the foraging trail to the kinetic energy of the ants. The model predicts and the measurements support the existence of two distinct phase velocities.
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1. Introduction

Ants form one of several insect societies that organize themselves without central control, and as such, provide rich theoretical ground for the exploration and understanding of complex collective behavior through the self-organization of simple organisms. One distinct advantage to studying ants is that careful experiments are inexpensive to implement and relatively easy to perform. Another is that there is a wealth of observational data on many ant species providing considerable insight into individual ant behavior and the various modes of local communication that facilitate colony behavior (Hölldobler and Wilson, 1990, 1994; Wilson, 1962). Ants within a colony communicate locally through chemical pheromones that can alert the ants of enemies or guide them to the food source and back to the nest. More specifically, ants are able to communicate with one another through secretions, passing excitement or odors (Peggy and Larson, 1965). While the visual acuity of ant varies from species to species, chemical trail laying and trail following are the dominant processes in finding the transporting food to the nest. If the food source is moved from its original location after a pheromone trail is developed, ants will continue to follow the trail even though it no longer leads to food. However, trail laid by many species of ants have short lifetimes (e.g., roughly 6 min for *Pheidole* Goetsch (1957) or 2 min for *Solenopsis saevissima* Bossert and Wilson (1963)) and therefore must be sustained by a regular flow of foraging ants. By "regular flow," we do not imply that the flux of ants is constant. To the contrary, our measurements and model predictions indicate that this is not the case generically. Rather, a regular flow means that many ants traverse any given position on the trail during any 6 min interval. In addition, it has been observed through experimentation that some species of ants return to the nest following their own scent trail (Skaife, 1961). What fascinates most investigators is that there has been no observation or measurement of global control or coordination. Rather, the global behavior of the colony is a product of simple local interactions among equals or different castes within a colony. Therefore, parameters governing simple individual behavior impact the growth and development of the entire ant colony complex. Our ability to identify and understand the parameters driving these behaviors will help resolve fundamental questions of the duration, fate and long-term evolution of insect societies (Bonabeau et al., 1997; Schultz, 2000). Thus, we turn our attention to a detailed study of ant foraging trail behavior,

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so that we can deduce fundamental local parameters that govern global outcomes.

In most species of ants, including those used in the experiments described in this paper, vision is severely limited. However, they are sensitive to tactile cues from nearby ants and a wide variety of chemical signals deposited by other ants. While there have been experiments indicating that foraging ants of *Lasius niger* traveling from a nest toward food possess little or no memory of past trail selection (Beckers et al., 1992b), there have been studies indicating that ants perform rudimentary path integration when returning to a nest site (Müller and Wehner, 1988). Cognition appears to be limited to the operation of different states, as if individuals switch from one basic algorithm to another from a limited selection of available algorithms. Examples include searching for food, transporting food or material to or from the nest or being alarmed. One of the simplest units of collective behavior is the foraging trail, a bi-directional flow of ants between a food source and the nest. The ant colony we study, *Tetramorium caespitum* (*L.*), exhibits the foraging behavior commonly seen on kitchen counters and picnic tables. The original aim of this project was to find parallels between ant-traffic patterns and human vehicular traffic, similar in spirit to the study by Burd et al. (2002) and John et al. (2004). In particular, we were interested in whether ants would resolve traffic paradoxes where humans behavior fails to achieve optimal configurations (see Arnott and Small, 1994 for some sample configurations). These traffic studies have at their core the assumption that there is a functional relationship between density and flow rate. A large number of observations and measurements led us to the conclusion that a strong functional relationship does not exist for our ant colony, even in well-established foraging trails (for example, look ahead to Fig. 3). Instead, we observe a rich spatio-temporal pattern of waves moving back and forth through the foraging trail. Since the foraging trail is the fundamental unit of organized behavior in ant colonies, it is crucial that it be studied systematically. This paper is an attempt to model ant foraging trails systematically and match model predictions with laboratory observations and measurements.

Leaving aside the spatio-temporal dynamics of the foraging trail, a large number of notable investigations have been dedicated to modeling ant colony behavior as a whole, or using general properties of ant foraging to develop useful algorithms. For instance, Dorigo et al. have developed an array of algorithms for solving the Traveling Salesman Problem, quadratic assignment and the job-shop scheduling problem (Bonabeau et al., 2000; Dorigo et al., 1996) with considerable success. Others have proposed using ant inspired algorithms for diverse applications such as robotic control (Zitar, 2005), managing peer to peer networks (Shen and Asano, 2003; Shen and Jaikaeo, 2005) and self-assembly using swarm robotics (Wilson et al., 2004).

Another line of investigation is to attempt to create artificial systems inspired by ant colonies and then understand their complexity (Chowdhury et al., 2002; Kluegl et al., 1998). Indeed, there is no dispute that very simple computational algorithms or physical mechanisms can produce very complex observable behavior, a theme at the core of Braitenberg's classic *Vehicles*, *Experiments in Synthetic Psychology* (Braitenberg, 1984). Thus, observing complex behavior from simple organisms such as ants, one can develop artificially simple structures and observe whether or not they produce complex behavior as a group, and if so, what sort of complex behaviors correspond to what features in the simple structures.

This paper introduces and analyses a simple model of an ant foraging trail with the hope that the analysis of the resulting system can inform us about real ant behavior and vice versa. Thus, the interplay between experimental observation and theoretical modeling play a central role in this paper. This investigation into spatio-temporal evolution of ant foraging trails was driven by experimental observations such as those shown in Fig. 1. In these space–time measurements, the foraging trail was observed for over an hour. Ant densities are observed to fluctuate in both space and time as waves, indicated by background arrows, traveling left and right. More details on the experimental procedures are contained in Appendix A. The ant foraging trail can be a very complex environment with many waves interacting with one another. Notice that the phase velocities are the same in both segments. Our goal is to develop a simple model with a small number of parameters that is capable of capturing the essentials of the observed ant behavior in our experiments. We find that this model predicts ant density waves along single foraging trails, and we will present experimental measurements of simple ant foraging trails which clearly exhibit traveling waves in agreement with the model. Furthermore, we will find that the reduced linearized model offers the advantage that it can be written entirely in terms of densities. One of the experimental difficulties with ant-traffic measurements is that it requires the measurement of ensemble ant velocities, a derivative field which by definition requires the spatial averaging of individual ant velocities. On the other hand, density is a macroscopic quantity that can be directly measured instantaneously. The model developed in this paper involves parameters which can be determined using density measurements exclusively.

This paper is organized as follows. In Section 1, we review the literature upon which this work is based and describe how this project relates to other investigations. In Section 2, we outline our fundamental assumptions and then derive a set of evolution equations for the densities of ants in an established foraging trail. In Section 3, we study the properties of the linearized evolution equations and determine and compare these results to measured quantities. In Appendix A, we provide the details of the ant colony experiments and measurements which inspired our modeling efforts and support the conclusions of our

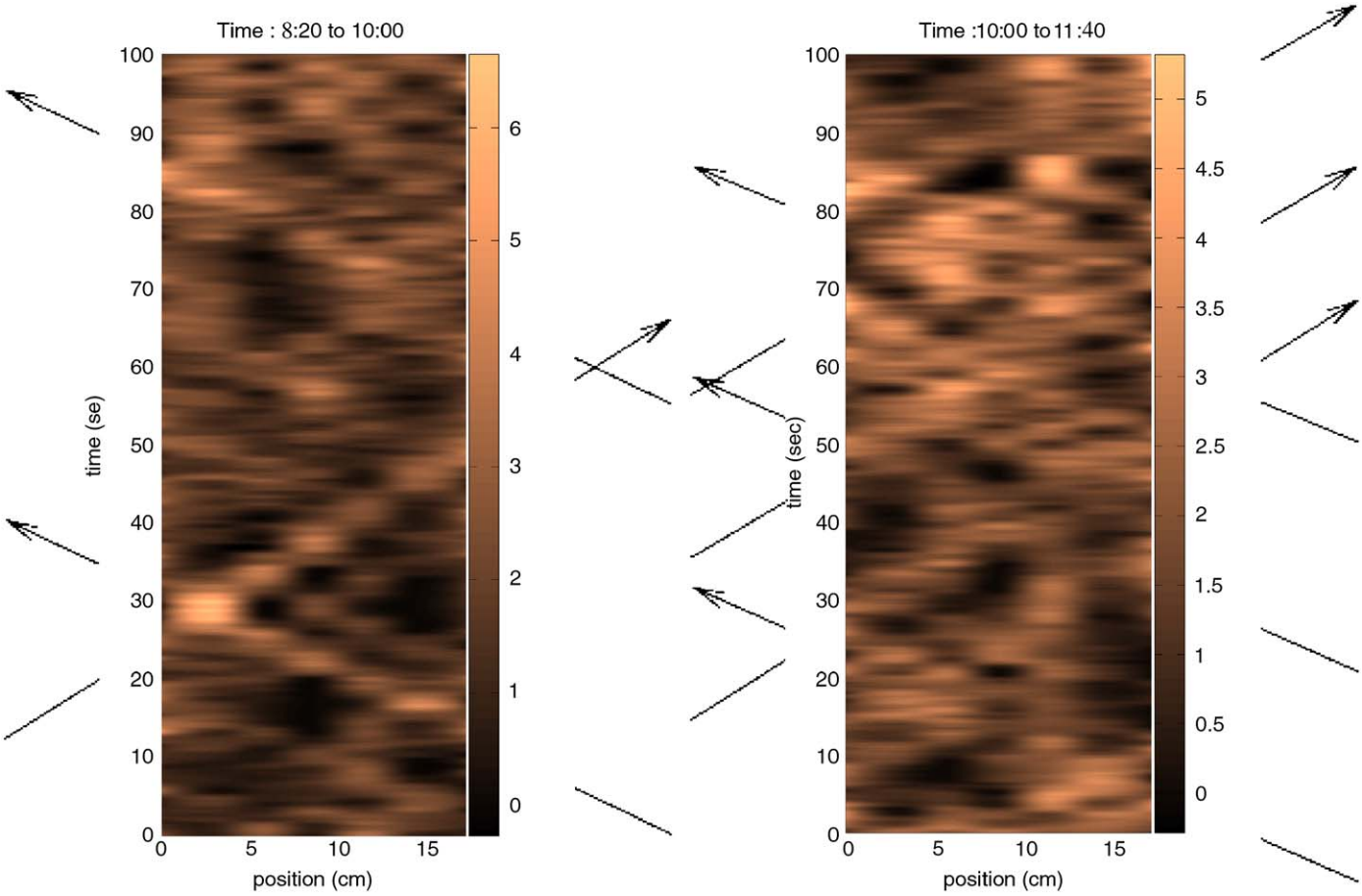

Fig. 1. Observations of ant foraging behavior. The ant density is measured in numbers of ants/cm along an ant foraging line. The ant nest is to the right of the domain over which measurements are made. The food is to the left of the measurement domain. At left, three high density waves are observed, two move to the left and one to the right. At right, many left- and right-moving waves are present.

mathematical analysis. In Section 4, we summarize our findings and lay out a plan for future investigations.

2. Modeling a foraging trail

2.1. Assumptions

Models of individual foraging trails vary considerably from one investigation to the next, and highlight important modeling issues. Discretization of time and space rapidly becomes an issue in all investigations, whether one is modeling the nest, a colony or a network of foraging trails. For instance, if one develops a cellular model for ant behavior, the cell size and time step become model parameters. To connect the model to actual ant behavior, one must refine the study to a point where the effect of the time step and the cell size are negligible in the process. Before we develop our model, we advance two facts.

(1) Ant pheromone trails evolve as a continuum process. That is, the mean free path of the pheromones is orders of magnitude smaller than the size of the ant that detects it. Ants returning from a food source to the nest will mark their path by depositing a continuous chemical trail on the ground (Hölldobler and Wilson, 1990). Generally, these pheromones are not very volatile and will remain in place for minutes (Bossert and Wilson, 1963).
(2) Ants are discrete objects. While ants have finite size and so occupy a finite area in a domain, one can describe a point in space as being either occupied by an ant or not.

The model proposed in this paper treats both the pheromone trail and the ants themselves as a continuum, and so we depart from fact (2). Thus, any results apply in a regime where ant densities are large, which is not always the case in our experiments. A number of behavioral assumptions lie at the heart of our modeling activities.

(1) We assume there are two modes of operation in the foraging trail, ants that are seeking food and ants that are transporting food to the nest. There is ample evidence that these two modes exist, along with many others in ant species (see Hölldobler and Wilson, 1990 for a survey). For instance, Kluegl et al. (1998) have developed a discrete space–time cellular automata

model that includes the effects of foraging, recruiting, energy storage, breeding and recruitment among others.

(2) We assume that ants communicate with other nearby ants through physical or chemical encounters. There are a number of modes of communication including transporting nest pheromone, laying trails via the ants' Dufour's gland, regurgitation and antennation (Beckers et al., 1992a, 1990; Burd and Aranwela, 2003; Wilson, 1962). With regard to pheromone trails and detection, considerable effort has been dedicated to understanding the physiology, transport and diffusive properties of the relevant compounds (for example Bossert and Wilson, 1963; Katazv-Gozansky et al., 2003). A common theme in most of these forms of communication is that ants moving in one direction along the foraging trail are passing information to ants moving in the opposing direction. This will be reflected in our ant behavior model.

(3) We assume no U-turning behavior on the foraging trail. Thus, ants seeking food continue until they find it, and ants transporting food continue until they reach the nest. U-turning behavior, studied in detail by Beckers et al. (1992b), may play an important role in recruitment and initial trail laying and development. We limited our study to well-established foraging trails, and we did not detect any U-turning behavior in our foraging trail measurements.

With these relatively minor assumptions, we have a durable framework for modeling a single foraging trail.

2.2. Quantification of foraging ants

To begin, we model two populations of ants, ants that are moving from the nest toward the food and ants that are moving from the food toward the nest. The first group of ants shall be referred to as population A. They shall have density $\rho_A(x,t)$ and travel with velocity $v_A(x,t)$. Similarly, the second group shall be referred to as population B with density and velocity $\rho_B(x,t)$ and $v_B(x,t)$, respectively. Each population has a density and speed governed by physical requirements and behavioral parameters. With four unknown functions, we seek a model for the foraging process that will involve four equations. While ant path integration is a fascinating topic all by itself (Couzin and Franks, 2003; Müller and Wehner, 1988), our experimental and mathematical domain is the simplest possible. The domain shall consist of a single one dimensional interval as shown in Fig. 2, and we assume that ants have no difficulty passing by one another as they travel back and forth.

To find a set of evolution equations for ρ_A, v_A, ρ_B and v_B, we first apply the basic conservation principle that ants are conserved. That is, ants are neither created nor destroyed. The flux of A and B ants from left to right is $\rho_A v_A$ and $\rho_B v_B$, respectively. Considering any interval $[a,b]$ of the domain, we can see that a difference in flux entering

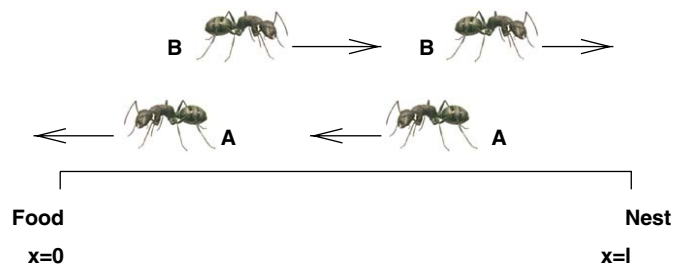

Fig. 2. Physical domain and configuration for the ant model. Population A ants leave the nest and seek food. Population B ants leave the food and seek the nest.

or leaving the interval must be balanced by a change in the total occupancy in that interval. The Fundamental Theorem of Calculus transforms this into a simple conservation principle for ants.

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_a^b \rho_A(x,t)\,\mathrm{d}x = \rho_A(b,t)v_A(b,t) - \rho_A(a,t)v_A(a,t),$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\int_a^b \rho_A(x,t)\,\mathrm{d}x = \int_a^b \frac{\partial}{\partial x}(\rho_A(x,t)v_A(x,t))\,\mathrm{d}x,$$

$$\int_a^b \frac{\partial}{\partial t}\rho_A(x,t)\,\mathrm{d}x = \int_a^b \frac{\partial}{\partial x}(\rho_A(x,t)v_A(x,t))\,\mathrm{d}x,$$

$$\int_a^b \left[\frac{\partial}{\partial t}\rho_A(x,t) + \frac{\partial}{\partial x}(\rho_A(x,t)v_A(x,t))\right]\mathrm{d}x = 0.$$

Since this is true for any interval $[a,b]$, under reasonable assumptions about the continuity of ρ and v, the integrand must be zero, and so we obtain at the following system of conservation equations.

$$\frac{\partial}{\partial t}\rho_A + \frac{\partial}{\partial x}(\rho_A v_A) = 0, \tag{1a}$$

$$\frac{\partial}{\partial t}\rho_B + \frac{\partial}{\partial x}(\rho_B v_B) = 0. \tag{1b}$$

This is analogous to continuity equations arising in the study of fluid dynamics, and this would be the natural limit of any cellular model as the cell density grows infinite.

To close the system, we model the behavior of the ants. Our simple model is based on momentum conservation. The momentum of a group of ants will be changed by a biochemical gradient or stress causing them to accelerate or decelerate. This is in contrast to the ant-traffic study by Burd et al. (2002), where one assumes that there is a functional relationship between the ant flow rate and the density: $q \equiv q(\rho)$. There are two reasons to look beyond this assumption. The first reason is that this formulation may lead to non-physical solutions such as rarefactions and shocks in ρ. Higher order vehicular and pedestrian traffic models remedy these deficiencies by including acceleration terms (for example, see Whitham, 1974 for discussion and analysis). The second reason is that the data in our own experiments and the experiments of Burd et al. indicate

that an explicit flow–density relationship does not explain all the data (see Burd et al., 2002). Burd et al. used a three parameter (v_0, ρ_0 and n) nonlinear fit of the form

$$v = v_0[1 - (\rho/\rho_0)^n] \tag{2}$$

to their data on *Atta cephalotes*. Results from our simple experiments on *T. caespitum*, shown in Fig. 3, reveal less of a correlation than that found by Burd et al. The experimental methods are explained in Appendix A. Linear regression yielded a coefficient of determination of 0.20. It is not surprising that our results are different from Burd et al.'s because measurements were taken in different settings with different species of ants. We obtained similar results to those shown in Fig. 3 in careful experiments where the width of the foraging trail was systematically constrained using petroleum jelly in an attempt to artificially raise the ant density. We succeeded in raising the density, but we did not succeed in controlling the ants' speed variations. Similar to the higher order terms incorporated by Whitham, we will model ant acceleration and deceleration terms through a stress equation. The resulting system of equations arising from our simple assumptions about ant behavior is somewhat unique.

With this reasoning, we aggregate a number of different physical processes that ants use to communicate. Ants transporting food communicate with foraging ants through pheromone secretion, regurgitation and sometimes auditory cues (Hölldobler and Wilson, 1990). Similar studies with leaf-cutting ants (*A. cephalotes*) have found that ants moving in opposite directions exchange leaf fragments and other information (Burd and Aranwela, 2003; Howard et al., 1996). We consolidate all of the short-range information exchanged between food-bound and nest-bound ants into distinct ant stresses. The ant stress can be interpreted as a simple chemical concentration, but it could include other effects as well. If we denote the ant stress for the nest-bound ants as $\sigma_N(x,t)$ and for the food-bound ants $\sigma_F(x,t)$,

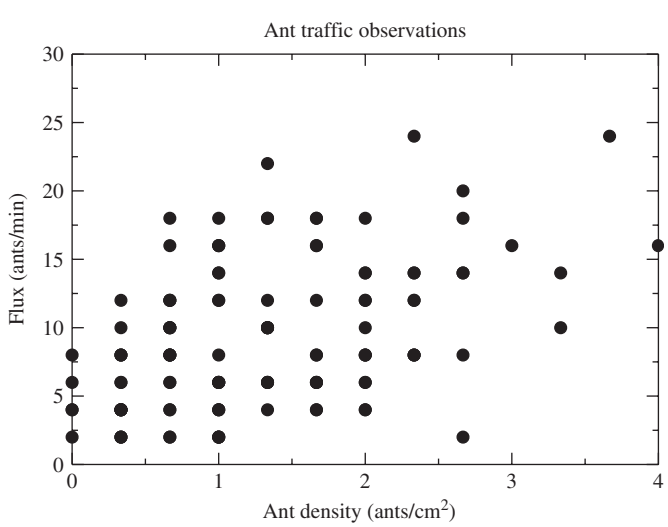

Fig. 3. Measurements of ant flow rate versus ant density.

then changes in σ_F will cause population A ants to accelerate or decelerate

$$\frac{\mathrm{d}}{\mathrm{d}t}(v_A) = \frac{\partial}{\partial x}\,\sigma_F(x,t), \tag{3a}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}(v_B) = \frac{\partial}{\partial x}\,\sigma_N(x,t). \tag{3b}$$

The left hand terms can be expanded further

$$\begin{aligned}\frac{\mathrm{d}}{\mathrm{d}t}v_A(x,t) &= \frac{\partial}{\partial t}\,v_A(x,t) + \frac{\mathrm{d}x}{\mathrm{d}t}\frac{\partial}{\partial x}\,v_A(x,t)\\ &= \frac{\partial}{\partial t}\,v_A(x,t) + v_A(x,t)\frac{\partial}{\partial x}\,v_A(x,t),\end{aligned}$$

and similarly for the v_B acceleration term. In the fluids literature, this is often referred to as a material or convective derivative since one is calculating a time derivative while following the material trajectory of an ant. The stress terms σ would correspond to the negative of a fluid pressure. Finally, we need to model σ_F and σ_N. For simplicity, we do not include pheromone history effects and assume that the ant stress is proportional to the ant density. It is well documented that ants returning to the nest from a food source will lay a pheromone trail (see, for example, Hölldobler and Wilson, 1990). We assume that A (food-bound) ants receive a cue from the B (nest-bound) ants proportional to their density. We also assume that A ants will follow other A ants, so we incorporate two terms in our stress model:

$$\sigma_F = k_A\rho_B + k_C\rho_A. \tag{4}$$

This is similar to the discrete model of Couzin et al. (2002) for animal groups except that this only incorporates a "zone of attraction" . It is important to note some differences between this model and the ant foraging trail model of Couzin and Franks (2003). The Couzin and Franks model is a two-dimensional constant-speed model where ants follow a pheromone trail but avoid one another. The proposed model is a one-dimensional variable speed system with two different interacting types of ants (food-seeking and nest-seeking). In the Couzin and Franks study, there is an avoidance term in the model so that colliding ants turn away from one another. In the one-dimensional problem, avoidance is not possible but the interaction of the two populations drives the dynamics of the system. The new system is a simple model indicating that the response of the A ants would be to accelerate in proportion to the number of B ants present in their vicinity because the B ants are reinforcing the pheromone trail. However, the nature of the cues that B (nest-bound) ants receive from A ants is more problematic to model. Studies have shown that returning ants can navigate back to their nests via some form of rudimentary path integration (Müller and Wehner, 1988). At the same time, ants move along bi-directional lanes along foraging trails, and there is evidence that some form of tactile communication or reinforcement occurs as ants pass one another (Burd and Aranwela, 2003; Couzin, 2005). Thus, we assume that some cues, however

weak, drive the B ants to accelerate and decelerate with a stress term. We also include the mutual attraction term described for σ_F

$$\sigma_N = k_B\rho_A + k_C\rho_B. \tag{5}$$

This model can also be interpreted to capture any of a number of local cues that the model ants pass along to one another including regurgitation, antennation or exchange of foreign matter such as leaf fragments that are density dependent. The colloquial analogy would be to consider a hungry graduate student wandering through the halls of a building. If this student sees others with donuts, the hungry student will quicken his pace toward those with food, correctly or incorrectly inferring that they have acquired what he desires. If number of students with donuts diminishes, he slows thinking he is on the wrong track.

Using the same technique as before, we conserve momentum noting that the only process that will change the momentum of a group of ants. Our simple behavioral model is

$$\frac{\partial}{\partial t}\, v_A(x,t) + v_A(x,t)\,\frac{\partial}{\partial x}\, v_A(x,t) = k_A\,\frac{\partial}{\partial x}\,\rho_B(x,t), \tag{6a}$$

$$\frac{\partial}{\partial t}\, v_B(x,t) + v_B(x,t)\frac{\partial}{\partial x}\, v_B(x,t) = k_B\frac{\partial}{\partial x}\,\rho_A(x,t). \tag{6b}$$

Thus, we hope to model an ant foraging trail with (1) and (6) along with suitable initial data and boundary conditions.

3. Results from a linear analysis of the ant foraging trail model

The ant foraging trail model (1), (6) from the previous section is a coupled system of nonlinear partial differential equations, and so finding large, useful families of exact solutions proves difficult. Certainly simulations are possible, but in this section, we seek to gain some physical insight in the roles of the model parameters, by studying a linearization of this system. Thus, we will assume that ant densities and velocities are constant, signified with a "0", with small spatial and temporal disturbances, denoted by "1":

$$\rho_A(x,t) = \rho_A^{(0)} + \varepsilon\rho_A^{(1)}(x,t), \tag{7a}$$

$$v_A(x,t) = v_A^{(0)} + \varepsilon v_A^{(1)}(x,t), \tag{7b}$$

$$\rho_B(x,t) = \rho_B^{(0)} + \varepsilon\rho_B^{(1)}(x,t), \tag{7c}$$

$$v_B(x,t) = v_B^{(0)} + \varepsilon v_B^{(1)}(x,t), \tag{7d}$$

where ε is a small parameter, and we could interpret the constant quantities as a mean density or velocity. If we substitute (7) into (1) and (6), collect all the terms at order ε and neglect higher order terms of order ε^2 as being so small as to be discarded, we arrive at the following coupled, linearized system:

$$\frac{\partial}{\partial t}\,\rho_A^{(1)} + \rho_A^{(0)}\,\frac{\partial}{\partial x}\,v_A^{(1)} + \left(\frac{\partial}{\partial x}\,\rho_A^{(1)}\right)v_A^{(0)} = 0, \tag{8a}$$

$$\frac{\partial}{\partial t}\,\rho_B^{(1)} + \rho_B^{(0)}\,\frac{\partial}{\partial x}\,v_B^{(1)} + \left(\frac{\partial}{\partial x}\,\rho_B^{(1)}\right)v_B^{(0)} = 0, \tag{8b}$$

$$\frac{\partial}{\partial t}\,v_A^{(1)} + v_A^{(0)}\,\frac{\partial}{\partial x}\,v_A^{(1)} = k_A\,\frac{\partial}{\partial x}\,\rho_B^{(1)} + k_C\,\frac{\partial}{\partial x}\,\rho_A^{(1)}, \tag{8c}$$

$$\frac{\partial}{\partial t}\,v_B^{(1)} + v_B^{(0)}\,\frac{\partial}{\partial x}\,v_B^{(1)} = k_B\,\frac{\partial}{\partial x}\,\rho_A^{(1)} + k_C\,\frac{\partial}{\partial x}\,\rho_B^{(1)}. \tag{8d}$$

For the duration of this paper, we will assume that nest-bound and food-bound ants move at the same speed (Skaife, 1961), so $V = -v_A^{(0)} = v_B^{(0)}$.

With (8a) and (8b), we can solve for $(\partial/\partial x)v_A^{(1)}$ and $(\partial/\partial x)v_B^{(1)}$ in terms of the densities alone. If we differentiate (8c) and (8d) with respect to time, and replace terms involving $(\partial/\partial x)\,v_A^{(1)}$ and $(\partial/\partial x)\,v_B^{(1)}$, we can reduce (8) to a second-order coupled system in density only

$$\frac{\partial^2}{\partial t^2}\,\rho_A^{(1)} - 2V\,\frac{\partial^2}{\partial x\partial t}\,\rho_A^{(1)} + (V^2 + k_C\rho_A^{(0)})\frac{\partial^2}{\partial x^2}\,\rho_A^{(1)} + k_A\rho_A^{(0)}\,\frac{\partial^2}{\partial x^2}\,\rho_B^{(1)} = 0, \tag{9a}$$

$$\frac{\partial^2}{\partial t^2}\,\rho_B^{(1)} + 2V\,\frac{\partial^2}{\partial x\partial t}\,\rho_B^{(1)} + (V^2 + k_C\rho_B^{(0)})\frac{\partial^2}{\partial x^2}\,\rho_B^{(1)} + k_B\rho_B^{(0)}\,\frac{\partial^2}{\partial x^2}\,\rho_A^{(1)} = 0. \tag{9b}$$

Now we examine the units of model quantities and the role they play in the dynamics of this system. Aside from the dependent space and time quantities, and the densities and velocities, we see that the k's have units of *energy per unit density*. Also, if we designate a timescale τ, we determine a natural length scale τV. If we rescale space and time with a simple change of variables from x to $\tilde{x}$ and t to $\tilde{t}$, where

$$\tilde{x} = x/\tau V, \quad \tilde{t} = t/\tau,$$

and then dropping the tildes, we see that the system depends upon only two dimensionless parameters:

$$\frac{\partial^2}{\partial t^2}\,\rho_A^{(1)} - 2\,\frac{\partial^2}{\partial x\partial t}\,\rho_A^{(1)} + (1 + \Pi_{CA})\frac{\partial^2}{\partial x^2}\,\rho_A^{(1)} + \Pi_A\frac{\partial^2}{\partial x^2}\,\rho_B^{(1)} = 0, \tag{10a}$$

$$\frac{\partial^2}{\partial t^2}\,\rho_B^{(1)} + 2\frac{\partial^2}{\partial x\partial t}\,\rho_B^{(1)} + (1 + \Pi_{CB})\frac{\partial^2}{\partial x^2}\,\rho_B^{(1)} + \Pi_B\frac{\partial^2}{\partial x^2}\,\rho_A^{(1)} = 0, \tag{10b}$$

where

$$\Pi_A = \frac{k_A\rho_A^{(0)}}{V^2}, \quad \Pi_B = \frac{k_B\rho_B^{(0)}}{V^2}, \tag{11a}$$

$$\Pi_{CA} = \frac{k_C \rho_A^{(0)}}{V^2}, \quad \Pi_{CB} = \frac{k_C \rho_B^{(0)}}{V^2}. \tag{11b}$$

In a well-developed foraging trail $\rho_A^0 = \rho_B^0$, so can set $\Pi_C = \Pi_{CA} = \Pi_{CB}$. The dimensionless quantities Π_A, Π_B and Π_C are *Froude numbers* for the ant system, representing the ratios of potential energy stored in the foraging trail via cues such as pheromone or physical contact to kinetic energy of the ants.

We can study the coupled system on a periodic domain, with period λ, as one might expect to observe in a very long foraging trail. In this case, we can express the ant densities as complex Fourier series

$$\rho_A^{(1)} = \sum_{n=-\infty}^{\infty} D_n(t) e^{in\pi x/\lambda}, \tag{12a}$$

$$\rho_B^{(1)} = \sum_{n=-\infty}^{\infty} E_n(t) e^{in\pi x/\lambda}. \tag{12a}$$

Since densities are real, we know that $D_n(t) = \overline{D_{-n}(t)}$ and $E_n(t) = \overline{E_{-n}(t)}$ where the overbar denotes complex conjugation (CC). The problem is linear, so it is enough to solve for D_n and E_n for a single mode $k = n\pi/\lambda$. A full solution would then be a superposition of such solutions as represented in (12). If we make the substitution,

$$\rho_A^{(1)} = D(t) e^{ikx} + \text{CC}, \tag{13a}$$

$$\rho_B^{(1)} = E(t) e^{ikx} + \text{CC}, \tag{13b}$$

in (10), we can reduce our coupled system of PDEs into a system of coupled ODEs

$$D'' - 2ikD' - k^2(1 + \Pi_C)D - k^2 \Pi_A E = 0, \tag{14a}$$

$$E'' + 2ikE' - k^2(1 + \Pi_C)E - k^2 \Pi_B D = 0. \tag{14b}$$

If we define

$$\vec{x} = \begin{bmatrix} D \\ E \\ D' \\ E' \end{bmatrix}, \quad M = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ k^2(1+\Pi_C) & k^2\Pi_A & 2ik & 0 \\ k^2\Pi_B & k^2(1+\Pi_C) & 0 & -2ik \end{bmatrix}, \tag{15}$$

then we can solve (14) by solving

$$\frac{d\vec{x}}{dt} = M\vec{x}. \tag{16}$$

The solution to this system is a superposition of four functions with the form $e^{ikx+\lambda t}$ where λ is an eigenvalue of M. Thus, the spatio-temporal dynamics of the linearization are determined by the eigenvalues of M which are

$$\lambda = \pm ik\sqrt{1 - \Pi_C \pm \sqrt{-4\Pi_C + \Pi_A \Pi_B}}. \tag{17}$$

There are four non-degenerate regimes possible.

(1) $\mathbf{\Pi_A \Pi_B < 4\Pi_C}$.
All four eigenvalues are complex with non-zero real parts. Two will have positive real part and two will have negative real part. The four fundamental solutions will correspond to left- or right-moving, exponentially growing or decaying traveling waves with a single phase velocity.

(2) $\mathbf{\Pi_A \Pi_B > 4\Pi_C}$.
In this case, three subcases are possible.

(a) $\mathbf{1 - \Pi_C + \sqrt{-4\Pi_C + \Pi_A \Pi_B} > 0}$ and $\mathbf{1 - \Pi_C + \sqrt{-4\Pi_C - \Pi_A \Pi_B} > 0}$.
The four solutions will be left- and right-moving traveling waves each with one of two possible distinct phase velocities. If we define the following:

$$c_1 = \sqrt{1 - \Pi_C + \sqrt{-4\Pi_C + \Pi_A \Pi_B}}, \tag{18a}$$

$$c_3 = \sqrt{1 - \Pi_C - \sqrt{-4\Pi_C + \Pi_A \Pi_B}}, \tag{18b}$$

we can write the full solution to (14) in terms of (15) as

$$\vec{x}(t) = a_1 \vec{v}_1 e^{ik(x - c_1 t)} + a_2 \vec{v}_2 e^{ik(x + c_1 t)} + a_3 \vec{v}_3 e^{ik(x - c_3 t)} + a_4 \vec{v}_4 e^{ik(x + c_3 t)} + \text{CC}, \tag{19}$$

where the a's are constants determined by the initial conditions and the $\vec{v}$'s are eigenvectors corresponding to the individual eigenvalues of M.

(b) $\mathbf{1 - \Pi_C + \sqrt{-4\Pi_C + \Pi_A \Pi_B} > 0}$ and $\mathbf{1 - \Pi_C - \sqrt{-4\Pi_C - \Pi_A \Pi_B} < 0}$.
Two solutions will be left- and right-moving traveling waves with a single phase velocity, and two will be solutions corresponding to exponential growth and decay.

(c) $\mathbf{1 - \Pi_C + \sqrt{-4\Pi_C + \Pi_A \Pi_B} < 0}$ and $\mathbf{1 - \Pi_C - \sqrt{-4\Pi_C - \Pi_A \Pi_B} < 0}$.
Two solutions will correspond to exponential growth, and two will correspond to exponential decay.

It is important to note that the analysis cannot explain how a particular disturbance arose in the system. Measurements show sustained spatio-temporal fluctuations, but the origins remain a mystery. They could be driven by boundary conditions, or nonlinear saturation of the exponential growth terms.

Now we seek to understand how the model can explain the data as observed and measured. The data clearly shows density disturbances traveling left and right. When many of these waves are superposed such as in Fig. 1 (right), modulations are likely to obscure direct observation of phase velocities. Fig. 4 provides more evidence of the two populations of ants driving one another. We see that A and B ants are moving at two distinct phase velocities, but the two wave trains in this series are synchronized in the sense that the temporal separation between to the two left-moving waves (40 s) is the same as the temporal separation between right-moving waves. One of the shortcomings of our experiments is that we are not able to separate ρ_A from ρ_B in our acquisition, so we cannot directly fit values of

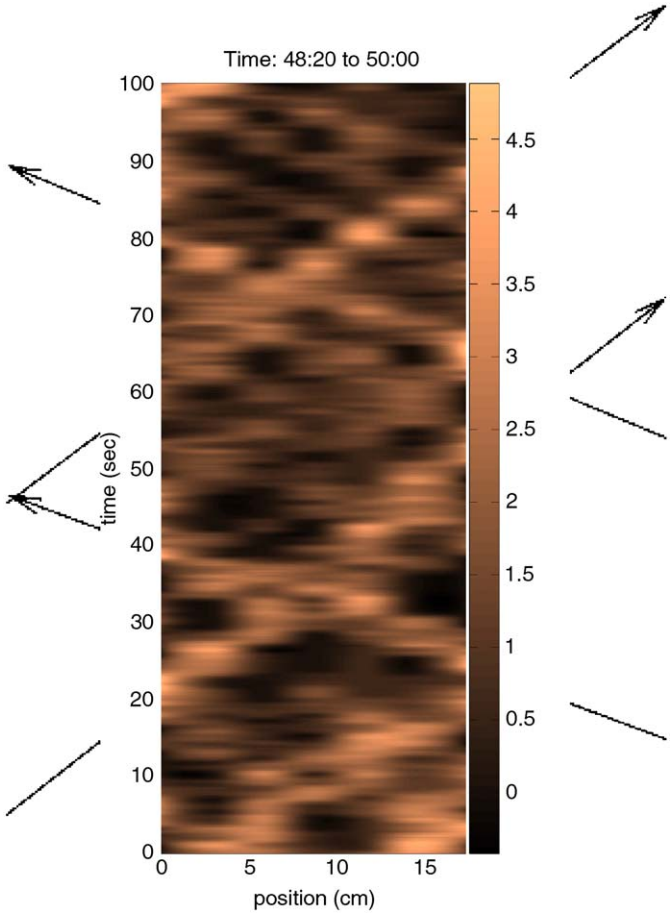

Fig. 4. Isolated waves traveling at different speeds.

Π_A, Π_B and Π_C, but in Fig. 4 we can actually see two distinct populations separated from one another for brief periods. The observations help us understand which regime explains the measured data. We discuss the prospects in the order they were presented.

(1) This regime does not explain all the observed data because it only predicts a single phase velocity when two are clearly observed.
(2) (a) This regime explains the observations. Two distinct phase velocities are observed without growth or decay.
    (b) This regime does not explain the fact that two phase velocities are observed as shown in Fig. 4.
    (c) This regime does not produce any traveling waves, so does not explain the flow of density variations in the system.

To conclude, only (2a) is supported by observation and measurement. Since B ants are capable of path integration independent of any input from A ants, studying whether or not populations of A ants provide any cues at all to B ants remains an area of active investigation. In this model, if the A ants provided no cues to B ants, this means that Π_B would be zero, making regime (2a) impossible.

4. Conclusions and future work

In this paper, we have presented controlled experiments of single ant foraging trails, and we observe complex patterns of traveling waves consisting of food-bound and nest-bound ants. These measurements show little direct correlation between speed and density in the ant foraging trails, as is often assumed in low order traffic models. Based on these observations and the work of others, we have systematically derived a continuum model for the evolution of ant densities under reasonable assumptions. Using a simple expression for ant stress, we find that the entire system can be reduced to three Froude numbers quantifying the coupling between nest-bound and food-bound ants. We have found that the linearized model has four regimes, but only one of them is supported by observations and measurements of traveling waves with two distinct phase velocities, corresponding to fast and slow moving waves. The linearized model offers the added advantage that the system can be completely expressed in terms of density so ensemble velocity measurements are not required.

There are a variety of directions to follow from this point, and we hope to continue along these lines of investigation. One of the fundamental biological questions that we have not fully investigated is fitness. Do the density variations contribute to or detract from fitness? If so, in what way and are there optimal combinations of the Froude numbers? If not, are the variations an artifact of some other physical and behavioral feature? Also, the model can be refined in many ways. The ant stresses do not include history effects, but chemical markers and other information can last for minutes along a foraging trail. One could and should include history effects in the ant stress terms. Appropriate boundary conditions are far from clear for ant foraging trails. Non-reflecting conditions are appropriate for population B ants returning to the nest, and specifying an outward flux of A ants from the nest is reasonable. However, ants swarm around the food, and modeling the interplay between arriving A ants and departing B ants will prove challenging. Also, there are many directions in which one could proceed to refine the ant stress model. For instance, searching behavior, a transient rather than periodic effect, could be included by adding a small stochastic term. While the linearization of the problem is beneficial for gaining physical insight into the essential mechanisms in the problem, full nonlinear calculations would permit the modeling of more complex behavior.

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Appendix A. Experimental methods

A.1. Procedures

The ant colony we study, *T. caespitum*, is kept in an ant table where we also conduct experiments. The entire ant table is 92 cm × 60 cm × 5 cm with wood sides and bottom and a removable plexiglass top secured with bolts and nuts. The table is split into two different subsections; the habitat side and the experimental side. The experimental side of the table is a flat wood surface, while the habitat side has a Sterilite plastic container (36 cm × 21 cm × 12 cm) secured in the middle of the wood bottom. The plastic container is filled with the original dirt of the ant nest and the nest itself. The population size of the ant we colony we study is unknown; however, we estimate 200 workers to be on the surface of the ant table at one time. In the experiments we conducted, the ants were fed a tablespoon of ant food inspired by the Bhatkar Diet, after being starved for three days (Hölldobler and Wilson, 1990). The ant food is made by combining 1 cup of boiling water with Knox unflavored gelatin, 1 egg, 1 tablespoon of honey, $\frac{1}{4}$ teaspoon of shaved multivitamin, and $\frac{1}{4}$ teaspoon of table salt. One tablespoon of ant food was placed on the experimental side of the ant table allowing the foraging trail from the ant nest to be 29 cm long. One hour after the food was placed on the ant table, the ants had found the food and a foraging trail had formed.

A.2. Data collection

The foraging trail was videotaped for one hour using the Canon Fluorite 100 × 3CCD video camera. The video was then loaded into the computer and saved capturing five frames/s. The original size of the images were 320 by 240 pixels.

We collected 18,000 frames in the series used for this paper. We apply a binary threshold to each bitmapped frame so that pixels are either black or white, and ants would appear as a certain number of black pixels. Thus, we can "count" the number of ants in an image or any region within an image by counting the number of black pixels. The digitalized frame was cropped to include only the foraging trail at 120 by 280 pixels. As shown in Fig. A.1, the fraction of pixels occupied by ants is directly proportional to the number of ants in the image. To certify that the computed pixel densities were an accurate representation of the number density of ants along the foraging line, we compared ant occupancy levels manually with computed pixel densities over a random sample of frames. We obtained a linear correspondence as we hoped. The foraging trail is divided into six segments for consideration. No systematic aberrations indicative of spatial bias were observed, so we see that pixel densities in foraging trail segments is an accurate representation of the number ants present in a segment of a foraging trail.

Ants densities were calculated over subintervals of the full foraging line. The measurements presented in this paper were made over seven subintervals, each having a length of 2.89 cm. Since ants only occupy a small number of pixels owing to the resolution of our camera, slight movements from one frame to another creates slight density variations from frame to frame. In order to smooth out these variations, we used a moving average when the images were processed to calculate the density. Another interpretation of this procedure is that our ρ can be interpreted as the probability that ants are occupying that space rather than being the exact density of ants in that specific time and space. If d_i^k is the pixel density of subinterval i at frame k, $\bar{d}_i^k$ is the average pixel density in subinterval i over m frames

$$\bar{d}_i^k = \frac{1}{m}\sum_{n=0}^{m-1} \mathrm{d}_i^{n+k}. \tag{A1}$$

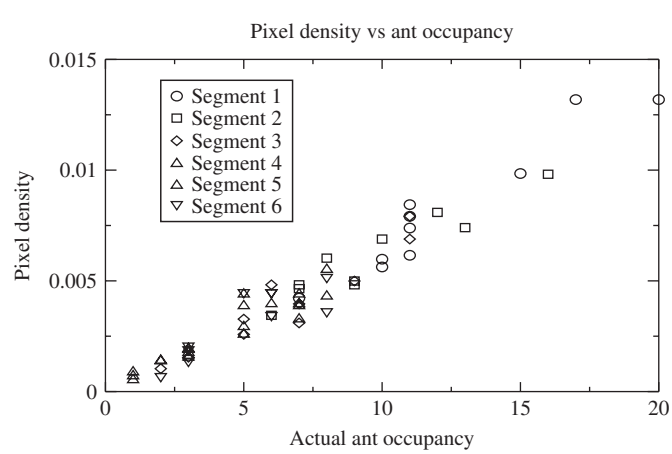

Fig. A.1. A correlation plot of ant densities counted manually versus pixel density.

In our case, we average over $m = 3$ frames. Ants in the foraging trail travel at about 1 cm/s, thus it takes an ant roughly 2.9 s to traverse one subinterval. Our averaging period is 0.6 s, so we are eliminating noise while still accurately tracking the ant density evolution.

To measure the flow–density relationship shown in Fig. 3, six 3 cm long by 1 cm wide subregions along the foraging line were marked for measurements. Measurements were made in 30 s intervals from video recordings. Occupancy data was gathered instantaneously by manually counting the ants in a subsection of the foraging trail at middle of each 30 s interval. Flow rate in units of ants/min was measured by counting the number of ants crossing through the midline of each subsection during the 30 s interval. In a few rare instances, the occupancy was zero at the precise moment it was measured even though the average flow rate was non-zero over the duration of the 30 s interval.

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